Chapter 3: Equations and inequalities

Chapter 3 · Original slide 1 · Y1 Chapter 3 Equations and Inequalities

Y1 Chapter 3
Equations and Inequalities

Chapter 3 · Original slide 2 · Chapter Overview

There is little new content in this chapter since GCSE.

2: Simultaneous Equations using Graphs

Find the points of intersection of
$y = 3x^2 - 2x + 4$ and $7x + y + 3 = 0$

3: Solving Inequalities

4: Sketching Inequalities

1: Simultaneous Equations

Solve:

$x + y = 11$
$xy = 30$

Find the set of values of x for which:

$x^2 - 11x + 24 < 0$

Sketch the region that satisfies the inequalities:

$2y + x < 14$
$y \geq x^2 - 3x - 4$

Chapter 3: Equations and inequalities

Chapter 3 · Original slide 3 · Solutions sets

The solution(s) to an equation may be:

$2x + 1 = 5$

$x^2 + 3x + 2 = 0$

$x > 3$

$x^2 = -1$

$x^2 + x = x\,(x + 1)$

! The solutions to an equation are known as the solution set.

A single value:

Multiple values:

An infinitely large set of values:

No (real) values!

Every value!

The point is that you shouldn't think of the solution to an equation/inequality as an 'answer', but a set of values, which might just be a set of 1 value (known as a singleton set), a set of no values (i.e. the empty set $\varnothing$), or an infinite set (in the last example above, this was $\mathbb{R}$)

Chapter 3 · Original slide 4 · Solutions sets

Scenario	Example	Solution Set
	$x + y = 9$ $x - y = 1$	
No solutions:	$x + y = 1$ $x + y = 3$	
	$x + y = 1$ $2x + 2y = 2$	

For simultaneous equations, the same is true, except each 'solution' in the solution set is an assignment to multiple variables.

All equations have to be satisfied at the same time, i.e. 'simultaneously'.

A single solution:

$x^2 + y^2 = 10$
$x + y = 4$

Two solutions:

Infinitely large set of solutions:

Chapter 3: Equations and inequalities

Chapter 3 · Original slide 5 · 1 :: Simultaneous Equations

Recap!

Solve the simultaneous equations:

$x + 2y = 3$
$x^2 + 3xy = 10$

We can either use substitution (i.e. making x or y the subject of one equation, and substituting it into the other) or elimination, but the latter is easier for linear equations.

Solve the simultaneous equations:

$3x + y = 8$
$2x - 3y = 9$

We can't use elimination this time as nothing would cancel.

We instead:

(1) Rearrange linear equation to make x or y the subject.

(2) Substitute into quadratic equation and solve.

Chapter 3 · Original slide 6 · Test Your Understanding

Solve the simultaneous equations:

$3x^2 + y^2 = 21$
$y = x + 1$

Chapter 3 · Original slide 7 · Exercise 3A/B – Y1 Pages 40, 41

4 Solve the simultaneous equations:
$x + y = 3, \quad x^2 - 3y = 1$

Chapter 3 · Original slide 8 · Exercise 3A/B – Y1 Pages 40, 41

5 a By eliminating y from the equations $y = 2 - 4x, \quad 3x^2 + xy + 11 = 0$, show that $x^2 - 2x - 11 = 0$.
b Hence, or otherwise, solve the simultaneous equations $y = 2 - 4x, \quad 3x^2 + xy + 11 = 0$, giving your answers in the form $a \pm b\sqrt{3}$, where a,b are integers.

Chapter 3 · Original slide 9 · Exercise 3A/B – Y1 Pages 40, 41

b Find the second pair of solutions for the simultaneous equations.

Chapter 3 · Original slide 10 · Extension

1

[MAT 2012 1G] There are *positive* real numbers x and y which solve the equations

$2x + ky = 4,$
$x + y = k$

for:

A) All values of k;

B) No values of k;

C) $k = 2$ only;

D) Only $k > -2$

2

[STEP 2010 Q1] Given that

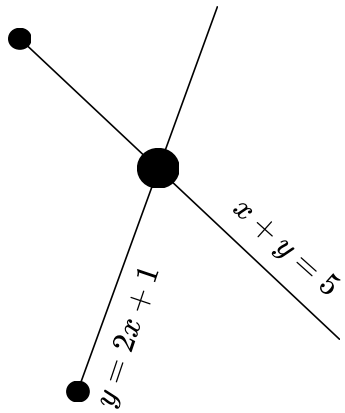
$5x^2 + 2y^2 - 6xy + 4x - 4y \equiv a(x - y + 2)^2 + b(cx + y)^2 + d$

a) Find the values of a, b, c, d.

b) Solve the simultaneous equations:

$5x^2 + 2y^2 - 6xy + 4x - 4y = 9,$
$6x^2 + 3y^2 - 8xy + 8x - 8y = 14$
(Hint: Can we use the same method in (a) to rewrite the second equation?)

Chapter 3 · Original slide 11 · Simultaneous Equations and Graphs



Now suppose we introduced a second simultaneous equation:

$y = 2x + 1$

$x + y = 5$

The second line again consists of all points (x,y) which satisfy the equation. So what point must satisfy both equations simultaneously?

Recall that a line with a given equation is the set of all points that satisfy the equation.

i.e. It is a graphical representation of the solution set where each (x, y) point represents each of the solutions x and y to the equation. e.g. $x = 3, y = 7$

Chapter 3 · Original slide 12 · Example

a) On the same axes, draw the graphs of:

$2x + y = 3$
$y = x^2 - 3x + 1$

b) Use your graph to write down the solutions to the simultaneous equations.

(We could always substitute into the original equations to check they work)

c) What algebraic method (perhaps thinking about the previous chapter), could we have used to show the graphs would have intersected twice?

Chapter 3 · Original slide 13 · Another Example

a) On the same axes, draw the graphs of:

$y = 2x - 2$
$y = x^2 + 4x + 1$

b) Prove algebraically that the lines never meet.

Chapter 3 · Original slide 14 · Final Example

a b

The line with equation $y = 2x + 1$ meets the curve with equation

$kx^2 + 2y + (k - 2) = 0$ at exactly one point. Given that k is a positive constant:

a) Find the value of k.

b) For this value of k, find the coordinates of this point of intersection.

Chapter 3 · Original slide 15 · Exercise 3C – Y1 Page 45

8 Given the simultaneous equations $2x - y = 1, \quad x^2 + 4ky + 5k = 0$, where k is a non-zero constant,

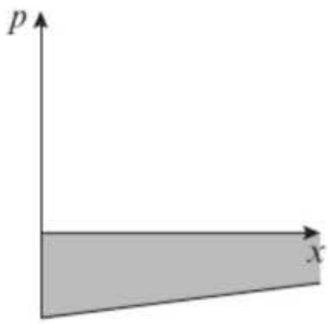
a show that $x^2 + 8kx + k = 0$.

Given that $x^2 + 8kx + k = 0$ has equal roots,

b find the value of k

c for this value of k, find the solution of the simultaneous equations.

Chapter 3 · Original slide 16 · Exercise 3C – Y1 Page 45



(5 marks)

9 A swimmer dives into a pool. Her position, p m, underwater can be modelled in relation to her horizontal distance, x m, from the point she entered the water as a quadratic equation $p = \frac{1}{2}x^2 - 3x$. The position of the bottom of the pool can be modelled by the linear equation $p = 0.3x - 6$. Determine whether this model predicts that the swimmer will touch the bottom of the pool.

Chapter 3 · Original slide 17 · Set Builder Notation

Recall that a set is a collection of values such that:

a) The order of values does not matter.

b) There are no duplicates.

Recap from GCSE:

• We use curly braces to list the values in a set, e.g. $A = \{1, 4, 6, 7\}$

• If A and B are sets then $A \cap B$ is the intersection of A and B, giving a set which has the elements in A and B.

• $A \cup B$ is the union of A and B, giving a set which has the elements in A or in B.

• $\varnothing$ is the empty set, i.e. the set with nothing in it.

• Sets can also be infinitely large. $\mathbb{N}$ is the set of natural numbers (all positive integers), $\mathbb{Z}$ is the set of all integers (including negative numbers and 0) and $\mathbb{R}$ is the set of all real numbers (including all possible decimals).

• We write $x \in A$ to mean "x is a member of the set A". So $x \in \mathbb{R}$ would mean "x is a real number".

$\{1, 2, 3\} \cap \{3, 4, 5\} =$

$\{1, 2, 3\} \cup \{3, 4, 5\} =$

$\{1, 2\} \cap \{3, 4\} =$

Chapter 3 · Original slide 18 · Set Builder Notation

It is possible to construct sets without having to explicitly list its values. We use:

The | or : means "such that".

$\{expr | condition\}$

or $\{expr : condition\}$

Can you guess what sets the following give?

$\{2x : x \in \mathbb{Z}\} =$

$\{2^x : x \in \mathbb{N}\} =$

$\{xy : x, y \text{ are prime}\} =$

We previously talked about 'solutions sets', so set builder notation is very useful for specifying the set of solutions!

Chapter 3: Equations and inequalities

Chapter 3 · Original slide 19 · Set Builder Notation

Can you use set builder notation to specify the following sets?

All (real) numbers greater than 5.

All (real) numbers between 5 and 7 inclusive.

All odd numbers.

All (real) numbers less than 5 or greater than 7.

Chapter 3 · Original slide 20 · Recap of linear inequalities

Inequality	Solution Set
$2x + 1 > 5$	
$-x \geq 2$	

Note: Multiplying or both sides of an inequality by a negative number reverses the direction.

If $x < 3$ and $2 \leq x < 4$, what is the combined solution set?

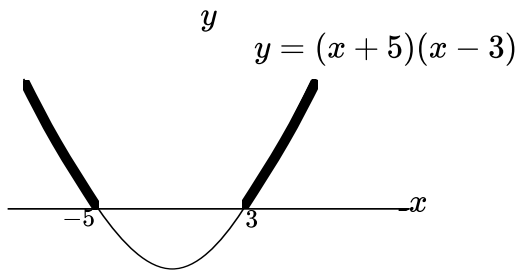
$3(x - 5) \geq 5 - 2(x - 8)$

Combining Inequalities:

If both inequalities have to be satisfied, we have to be on both lines. Place your finger vertically and scan across.

Chapter 3: Equations and inequalities

Chapter 3 · Original slide 21 · RECAP :: Solving Quadratic Inequalities



Solve $x^2 + 2x - 15 > 0$

Step 1: Get 0 on one side

$(x+5)(x-3) > 0$

Step 2: Factorise

Step 3: Sketch and reason

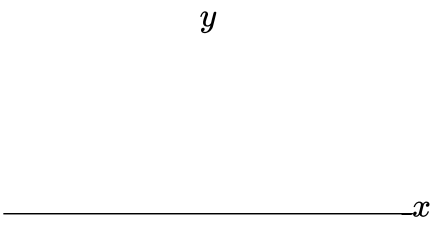
Since we sketched $y = (x+5)(x-3)$
we're interested where $y > 0$, i.e. the parts of the line where the y value is positive.

Click to bolden >

What can you say about the x values of points in this region?

What can you say about the x values of points in this region?

Chapter 3 · Original slide 22 · Solving Quadratic Inequalities



Solve $x^2 + 2x - 15 \leq 0$

Step 1: Get 0 on one side

$(x+5)(x-3) \leq 0$

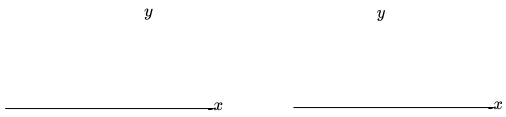
Step 2: Factorise

Step 3: Sketch and reason

Again, what can we say about the x value of any point in this region?

Note: we need $\leq$ rather than $<$ to be consistent with the original inequality.

Chapter 3 · Original slide 23 · Further Examples



Solve $x^2 + 5x \geq -4$

Solve $x^2 < 9$

Note: do not skip the 'sketch step'.

To use the quadratic inequality solver on ClassWiz, go to Menu → Inequalities, then choose 'order 2' (i.e. quadratic)

Try the same on a CG50.

Chapter 3 · Original slide 24 · Test Your Understanding

Find the set of values of x for which

(a) $3(x-2) < 8 - 2x$, (2)

(b) $(2x-7)(1+x) < 0$, (3)

(c) both $3(x-2) < 8 - 2x$ and $(2x-7)(1+x) < 0$. (1)

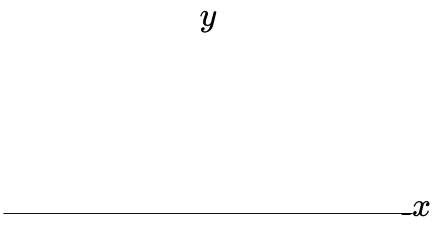
Note: the original equation has no solutions, but the inequality $q^2 + 8q < 0$ *did* have solutions. But we have found the solutions for q that result in the original equation not having any solutions for x. These are different variables, so have different solutions sets, even if the solution set of q influences the solution set of x.

Given that the equation $2qx^2 + qx - 1 = 0$, where q is a constant, has no real roots,

(a) show that $q^2 + 8q < 0$. (2)

(b) Hence find the set of possible values of q. (3)

Chapter 3 · Original slide 25 · Deal with inequalities with a division by x



Find the set of values for which $\frac{6}{x} > 2, x \neq 0$

Why can't we just multiply both sides by x?

Chapter 3 · Original slide 26 · Exercise 3D/3E – Y1 Page 47-48, 50-51

3 Use set notation to describe the set of values of x for which:

g $4(3x + 7) \leq 20$ or $2(3x - 5) \geq \frac{7-6x}{2}$

6 Find the set of values of x for which $x^2 - 5x - 14 > 0$.

Chapter 3 · Original slide 27 · Exercise 3D/3E – Y1 Page 47-48, 50-51

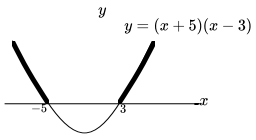
8 Given that $x \neq 3$, find the set of values for which $\frac{5}{x-3} < 2$.

Chapter 3 · Original slide 28 · Exercise 3D/3E – Y1 Page 47-48, 50-51

9 The equation $kx^2 - 2kx + 3 = 0$, where k is a constant, has no real roots. Prove that k satisfies the inequality $0 \leq k < 3$.

Chapter 3: Equations and inequalities

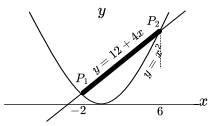
Chapter 3 · Original slide 29 · Inequalities on Graphs



When we solved quadratic inequalities, e.g.
$(x+5)(x-3)>0$

We plotted $y=(x+5)(x-3)$ and observed the values of x for which $y>0$.

Can we use a similar method when we don't have 0 on one side?



Example: L_1 has equation $y=12+4x$. L_2 has equation $y=x^2$.

The diagram shows a sketch of L_1 and L_2 on the same axes.

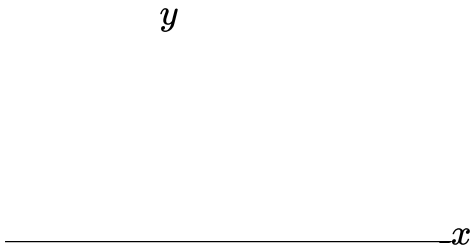
a) Find the coordinates of P_1 and P_2, the points of intersection.

b) Hence write down the solution to the inequality $12+4x>x^2$.

a b

Solve simultaneously to find points of intersection:

Chapter 3 · Original slide 30 · Inequality Regions



On graph paper, shade the region that satisfies the inequalities:

$2y+x<14$
$y\geq x^2-3x-4$

You did this at GCSE, the only difference here being that the graphs involved might not be straight lines.

Step 2:

An inequality involving x and y represents a 2D region in space.

Identify the correct side of each line each inequality represents.

Click to sketch >

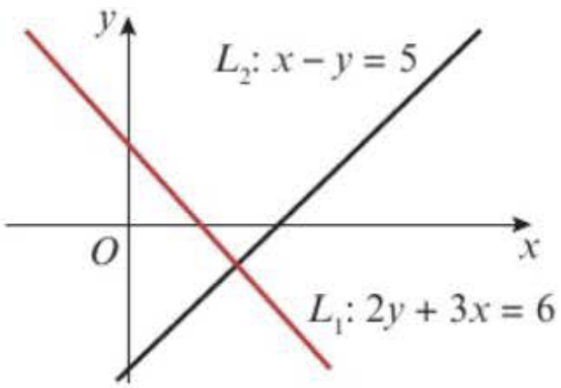
Tip: To quickly sketch $2y+x=14$, consider what happens when x is 0 and when y is 0.

Tip: Make sure y is on the side where it is positive.

If y is on the smaller side, you're below the line.

If y is on the greater side, you're above the line.

Chapter 3 · Original slide 31 · Exercise 3F/3G – Y1 Page 53, 55



1 L_1 has equation $2y + 3x = 6$. L_2 has the equation $x - y = 5$. The diagram shows a sketch of L_1, L_2.

a Find the coordinates of P, the point of intersection.

b Hence write down the solution to the inequality $2y + 3x > x - y$.

Chapter 3 · Original slide 32 · Exercise 3F/3G – Y1 Page 53, 55

3 Find the set of values of x for which the curve with equation $y = \mathrm{f}(x)$ is below the line with equation $y = \mathrm{g}(x)$.

a $\mathrm{f}(x) = 3x^2 - 2x - 1$, $\mathrm{g}(x) = x + 5$

b $\mathrm{f}(x) = 2x^2 - 4x + 1$, $\mathrm{g}(x) = 3x - 2$

c $\mathrm{f}(x) = 5x - 2x^2 - 4$, $\mathrm{g}(x) = -2x - 1$